

# 1. Core Approach: The Autonomous Spatial Commerce Platform

Bridging Generative AI with Deterministic Architectural & Financial Guardrails

| 0ms / 0 Tokens                     | 100% Exact                      | 15" & 21-30"                   | 100% Offline                  |
|------------------------------------|---------------------------------|--------------------------------|-------------------------------|
| Fast-Path Router Bypass            | Math & 18% GST Accuracy         | Building Code Clearances       | Edge Appliance Ready          |
| Instant greetings & catalog tables | Zero mental math hallucinations | Ergonomic fixture layout rules | Local Ollama + ChromaDB + BGE |

| THE LUXURY SPATIAL COMMERCE CHALLENGE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | OUR AGENTIC SOLUTION & CORE PARADIGM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• <b>Spatial Clearances Ignored:</b> Pure LLMs hallucinate fixture sizes, recommending 6ft tubs for 4x5ft powder rooms without verifying door swings or centerline codes.</li><li>• <b>Pricing &amp; Tax Hallucinations:</b> Multi-item luxury quotes with 15% bundle discounts and 18% GST in INR consistently fail standard arithmetic checks in generative LLMs.</li><li>• <b>High Latency &amp; Cloud Costs:</b> Traditional agent loops incur 3-6s delays and paid API tokens even for trivial product catalog requests and customer greetings.</li></ul> | <ul style="list-style-type: none"><li>• <b>Decoupled Dual-Engine Architecture:</b> Replaces open-ended chatbot prompts with a ReAct Orchestration Brain backed by 4 deterministic Python rule and math engines.</li><li>• <b>0ms Deterministic Intent Routing:</b> Regex classifier instantly intercepts greetings and catalog requests with zero LLM tokens and zero latency overhead.</li><li>• <b>Market-Specific Precision (India INR):</b> Native 18% GST tax rollups, PIN code delivery lead times, and localized rupee currency formatting across all touchpoints.</li></ul> |

## 2. System Architecture & End-to-End ReAct Workflow

Multi-Stage Orchestration: Fast-Path Intent Router -> Autonomous Agent -> Deterministic Rule Engines

|                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><b>1. Spatial Studio Engine</b><br/>tools/room_planner.py<br/>tools/bundle_optimizer.py<ul style="list-style-type: none"><li>Enforces 15" toilet centerline &amp; 21"-30" front clearance.</li><li>Calculates fixture capacities (compact powder room to master bath).</li><li>Automated theme bundles.</li></ul></div> | <div><b>2. 2-Stage Hybrid RAG</b><br/>tools/catalog.py<br/>chroma_db + cross-encoder<ul style="list-style-type: none"><li>Stage 1: Dense BGE-Small vector retrieval over ChromaDB.</li><li>Stage 2: Cross-Encoder MiniLM reranker for top-tier relevance.</li><li>In-memory LRU caching.</li></ul></div>     |
| <div><b>3. Precision Pricing Engine</b><br/>tools/calculator.py<br/>Strict Python Arithmetic<ul style="list-style-type: none"><li>Itemized quotes in INR.</li><li>15% automated package discount savings.</li><li>Exact 18% GST tax calculation with budget variance tracking.</li></ul></div>                               | <div><b>4. Logistics &amp; Stock Engine</b><br/>tools/inventory.py<br/>Regional Warehouses<ul style="list-style-type: none"><li>Real-time stock status (In Stock / Low Stock).</li><li>PIN-code based dispatch routing (Mumbai / Delhi / Bengaluru).</li><li>Verified 2-7 day transit times.</li></ul></div> |

**Zero-Lag Resiliency & Cloud Timeout Fallback:** If cloud API latency exceeds 3.0s or rate limits trigger (HTTP 429), the system deterministically synthesizes structured sales proposals from verified tool outputs in <50ms. Zero UI freeze, 100% uptime.

### 3. Technical Stack & Dual-Runtime Architecture

Engineered for Zero-Cost Edge Hardware or Scalable Multi-Tenant Cloud Deployment

| Architecture Layer     | Technology & Component                                    | Key Technical Capabilities & Rationale                                                                   |
|------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Agent Orchestrator     | LlamaIndex v0.11+<br>AgentWorkflow & FunctionCallingAgent | Event-driven ReAct loop managing stateful tool calls, intent routing, and dynamic context injection.     |
| Vector Database        | ChromaDB<br>chromadb.PersistentClient                     | Local embedded vector store with self-bootstrapping indexer (0-dependency cold startup).                 |
| Embeddings & Reranking | BGE-Small-v1.5 +<br>ms-marco-MiniLM-L-6-v2                | Local HuggingFace embeddings + Cross-Encoder reranker with in-memory LRU cache.                          |
| Dual LLM Runtime       | Ollama (Local) /<br>OpenRouter (Cloud)                    | 100% Offline via qwen2.5:3b (Flash Attention) with automatic fallback to cloud models.                   |
| Frontend Studio        | Gradio 6.0 Custom UI                                      | Interactive Spatial Studio 3D visualizer, responsive sliders, quick-action chips, and advisor chat.      |
| Packaging & CI/CD      | Docker Compose + One-Click Scripts                        | Production Docker containerization, 12-scenario automated verification suite, and Windows/Linux scripts. |

# 4. Innovation Pitch & Commercial Impact

Revolutionizing Luxury Sanitaryware Sales with Autonomous Spatial Intelligence

|                                                                                                                                                                                                                               |                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Hallucination-Proof Spatial CAD</b><br>Eliminates returns and architectural rework by deterministically validating room dimensions, minimum door swings (30"), and fixture centerlines (15") before proposing products. | <b>2. 0ms Zero-Token Instant Routing</b><br>80% of customer interactions (greetings, catalog exploration, single specs) bypass LLM inference entirely. Saves 100% token costs and provides instantaneous UI response. |
| <b>3. 100% Offline Edge Showrooms</b><br>Runs autonomously on in-store hardware via quantized Ollama weights without internet connectivity. Protects customer data privacy and operates with zero recurring cloud bills.      | <b>4. Complete INR Commercial Engine</b><br>End-to-end luxury sales workflow: automated 15% suite package discounts, 18% GST tax computations, and PIN code delivery lead times in a single unified proposal.         |

The Kohler  
Guarantees  
sharing (